

Oracle Label Security™ — Controlling Access to Data

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EXECUTIVE OVERVIEW

The need to assign sensitivity labels to data has existed for hundreds of years and continues to exist today in virtually every industry. Scientific discoveries made in the 21st century are no more closely guarded than scientific discoveries made in the 17th century. In 1998 Oracle added Virtual Private Database (VPD) to the security features available in Oracle Enterprise Edition. VPD greatly enhances the ability to control access to data by giving administrators the ability to add security policies to objects in the Oracle RDBMS. With VPD, application size and complexity decrease while security increases. This is because the VPD security policy is centralized in the Oracle server. Oracle Label Security TM is a powerful new security solution for controlling access to individual data rows and protecting critical data. Built on the same technology as VPD, it provides exciting new security functionality for industries where confidentiality, privacy and data sensitivity are vital to the success of the enterprise. Healthcare, financial services, law enforcement and defense are just a few of the industries which can benefit from Oracle Label Security.

INTRODUCTION

Restricting access to sensitive data

In a hosted application, what mechanism separates multiple companies sharing a single instance application? The answer is a sensitivity label. Sensitivity labels provide highly granular row level security capabilities. A sensitivity label can be as simple as a company name, or it can be a complex series of code words added to a document being delivered to the President of the United States or the Chief Executive Officer of a corporation. A sensitivity label can be used to restrict access to patient medical records or to protect an ongoing criminal investigation.

PROTECTING SENSITIVE DATA

Oracle Label Security is designed to address the sophisticated access control requirements found in today's e-business environment. It is the first label based access control product designed specifically for the most popular commercially available operating systems. Oracle Label Security enforcement options allow security policies to be finely tuned for specific industry environments.

Out-of-the-box VPD Security Policies

No software development necessary

The Virtual Private Database feature is an ideal solution for environments where access control is based on pure application data, and where PL/SQL development expertise exists. In such situations, security policies can be written using PL/SQL and assigned to tables and views using the VPD application programmatic interface (API). Oracle Label Security is a sophisticated out-of-the-box security solution designed specifically for label based access control. Oracle Label Security requires no software development. Oracle Label Security is an ideal solution where access control is based on the sensitivity of data and traditional access control mechanisms are not sufficient.

Advanced Row Level Security

Advanced row level security for highly granular access control

VPD was described in the previous section as being ideal for security policies where access control is based on pure application data. For example, imagine a listing which has three columns (*department number*, *department name* and *location*) and contains information on all departments in a corporation. A VPD security policy can easily be written which states that employees can only view information on the departments to which they are assigned. However, in other situations Oracle Label Security may be a better solution. What if an employee in a management role has responsibility over several departments? It would not make sense to restrict the manager from viewing the departments he or she manages. Secondly, what if the manager needs to view departments which he manages, but is restricted from making updates on some or all of the departments? Perhaps some departments should only be visible to the CEO? These are just a few examples of security policies which could be written using the VPD API, but which would require much more complex and costly software development. Oracle Label Security lets you avoid this overhead. Each of these examples can be handled out-of-the-box by Oracle Label Security with no software development. Easily customized sensitivity labels provide advanced row level security.

Sensitivity Labels

Sensitivity labels can be composed of a standalone level component, or a level component combined with compartments, groups or both. The ability to use all three components in a sensitivity label provides advanced row level security and a high degree of granular access control.

Example Sensitivity Labels:

Internal : Asia Production : Vice President

Judicial : National Security : White House

Sensitive : CDC : Atlanta

Confidential : Global Warming : United Nations

Levels – Internal, Confidential, Sensitive,
Highly Sensitive

Sensitivity Levels

One component of the data sensitivity label which Oracle Label Security supports is known as a level. The easiest way to understand levels is to use an example. The novel *The Hunt For Red October* by Tom Clancy, is based around a fictional super quiet Russian submarine. The existence of the submarine is an example of an extremely high sensitivity level, such as *Top Secret*. In this example, personnel whose authorizations do not meet or exceed the *Top Secret* authorization will be restricted from viewing details assigned to this sensitivity level.

- *Level* -- The level is a hierarchical component which denotes the sensitivity of the data. A typical government organization might define levels confidential, sensitive and highly sensitive. However, there is no requirement to define more than one level. For example, a commercial organization might define a single level for company confidential data or application hosting requirements.

Compartments – Company Separation,
Projects, Technology, Test Results

Sensitivity Compartments

Another component of the data sensitivity label which Oracle Label Security supports is known as a compartment or category. In *The Hunt For Red October* the water propulsion system is an example of a sensitive technology which might be assigned a security compartment. In this example, personnel whose responsibilities do not include the water propulsion system can be restricted from viewing details assigned the sensitivity compartment.

- *Compartment* - The compartment component is sometimes referred to as a “category” and is non-hierarchical. Typically one or more compartments are defined to segregate data. For example, a compartment might be defined for an ongoing strategic initiative, or might map to a hosted application subscriber. Users can be given read only or read-write authorizations on an individual compartment basis, providing powerful row level security.

Groups – Organizational ownership, finer granularity

Sensitivity groups

Another component of the data sensitivity label which Oracle Label Security supports is known as a group. In *The Hunt For Red October* the military organization owning the submarine is an example of a potential group. For example, imagine two groups called *General Operations* and *Special Operations* are defined in Oracle Label Security. The *General Operations* group is defined as a child of the *Special Operations* group. In this scenario, personnel authorized for the *General Operations* group will be restricted from viewing information assigned the *Special Operations* group. However, personnel authorized for the *Special Operations* group will be permitted to view information assigned either the *Special Operations* or *General operations* group. The *Special Operations* group provides a larger view of the submarine organization.

- **Group** - The group component is used to record ownership and can be used hierarchically. For example, two groups called *Senior VP* and *Manager* could be created and subsequently assigned as children of the *CEO* group, creating an ownership tree. Users can be given read only or read-write authorizations on an individual group basis, providing powerful row level security.

COTS solution – reduced development cost and risk

Reduced Development Cost and Risk Exposure

One of the key advantages Oracle Label Security offers is the fact that the software for row level security is provided out-of-the-box. It allows the developer or administrator to quickly and easily get started with sensitivity labels and highly granular row level security. In addition, Oracle Label Security enforcement options allow security to be tailored to the specific needs of the enterprise. With Oracle Label Security, no development costs are incurred writing the software to enforce row level security. Oracle Label Security can thus dramatically shorten the development timeline, resulting in lower development and maintenance costs. System integrators can bid Oracle Label Security as a commercial off the shelf (COTS) solution in proposals.

Performance – highly optimized algorithms

Optimized Performance

Row level security is, by its very nature, an intensive operation. Not only is an access control check performed at the database table level, but an additional check is performed for each individual data row stored in the database table. The processing required is directly related to the complexity of the security policy. Oracle Label Security has been highly optimized to process complex access control decisions based on data sensitivity labels. Using the C, PL/SQL and Java software development languages, the Oracle Label Security development team optimized the processing based on options selected by the security administrator. The Oracle Label Security development team continues to develop and refine sophisticated optimization routines for row level security.

**Designed for standard operating systems,
based on stringent government and
commercial requirements**

Operating System Independence and Real World Functionality

The evolution of the Internet and rapid consolidation within the computer industry has made flexibility a key factor in IT expenditures. Oracle Label Security is built to meet stringent, real world, operational requirements for row level security found in government and commercial organizations. Oracle Label Security is built for today's Internet environment and incorporates years of feedback received from Oracle customers in both government and commercial organizations. Oracle Label Security is available on standard, commercially available operating systems, giving IT managers, in government and commercial organizations, hardware flexibility.

Oracle Policy Manager GUI

Manage row level security with a mouse

Oracle Policy Manager is the GUI administration tool for Oracle Label Security. Based on the Oracle Enterprise Manager framework, Oracle Policy Manager can be used to define sensitivity labels, authorize users and protect tables or entire schemes in the Oracle database.

CONCLUSION

The Internet provides a labyrinth of pipelines to transmit information. However, in the rush to use the Internet to reduce costs, improve efficiency and speed communication, the information technology industry must provide information managers the ability to better manage access to critical data. Oracle Label Security meets this need by providing the ability to closely control access to data with a high degree of granularity. Utilizing data sensitivity labels plus an array of enforcement options, Oracle Label Security provides advanced row level security. Oracle Label Security enforces security within the Oracle database server, eliminating complex and costly application logic. Oracle Label Security is available for standard operating systems and requires no programming knowledge.



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